

02. Software Design - Requirements

The Story:

Clients request us to build software.

Then, how do we start?

Mr. Star's Email

- A business owner, Mr. Star, emails us.

Dear Python programmer,

My name is Mr. Star and I own Star Station, a full service gasoline station in Deer Creek, a small town in the middle of the United States.

The station has 7 employees and I use an excel sheet to keep track of their names and salaries. But we want to improve our administration, and I need some fancy business software. Could you build this for us?

The first version should just replace our Excel sheet and show a list of the employees with their salaries.

My nephew (who wants to be a programmer) started with a small script that shows all the employee names. Perhaps you can use it.

Requests

- The station has seven employees.
- He needs to keep track of names and salaries.

NAME	SALARY
Vera	2000
Chuck	1800
Samantha	1800
Roberto	2100
Joe	2000
Dave	2200
Tina	2300

Prototype

- In short, he wants to replace Excel sheets.
- His nephew (a wannabe software engineer) has already made a small script (prototype).

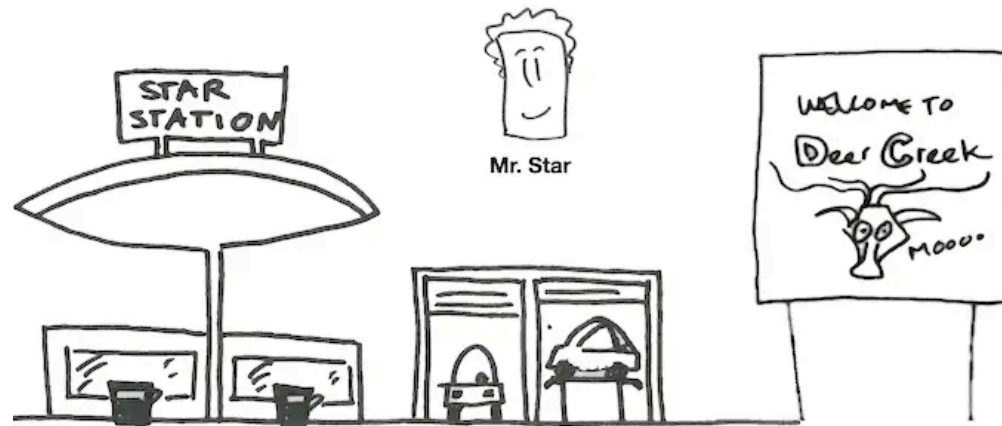
Before writing requirements, we must understand the business.

IMPORTANT

Business Logic

- **Business logic** (or **domain knowledge**) describes how the problem domain works.
- We must understand it before we can write useful requirements or design software.

- Here, the core business logic is tracking each employee and their salary.



The Request Details

- We have an Excel spreadsheet with all the information.

Employees



This is Vera. She is the manager at Star Station and the cashier



This is Roberto. He mans the kitchen and makes lovely sandwiches



Chuck and Samantha are the station attendants who fill up the gas tanks with gasoline.



This trio is Dave, Tina and Joe. Whatever is wrong with your car, they can fix it!

| In the Real-World Situation

- Clients often cannot fully describe what they need at the beginning.
- A prototype turns **unknown unknowns** into questions we can discuss.
- Feedback changes the requirements, so we should expect and design for change.

Business Logic in the Age of AI

- AI can generate code quickly, but it cannot compensate for a poorly understood problem.
- The engineer's focus therefore shifts toward business logic, requirements, design, and architecture.
- Starting without that understanding creates problems sooner or later.

We now understand enough of Mr. Star's business to write the first requirement.

IMPORTANT

Requirements

- Requirements are a set of descriptions of user requests.
- Requirements are the communication tool: clients and software engineers can agree upon the software to be delivered.

Warning: Bad format

Requirements Version 1

– The software shows a list of the employees with their salaries.

- According to this requirement, if we build software that shows a list of employees with their salaries, we will solve the problem.

User Stories

- However, the description is not accurate enough.
- We have a better way of writing down requirements with **user stories**.
 - It has the format: As a [WHO], I want [WHAT], so that [WHY].

- It identifies three pieces of information: **who** (line 1), **what** (line 2), and **why** (line 3) of the requirement.
- Now we are ready to start.

Requirements Version 2

1. As a "manager,"
2. I want to "see a list of the employees with their salaries,"
3. so that "I can easily understand what is going on."

Prototype (Version 0)

- Even though we have the requirements in the form of **User stories**, we still don't know what we don't know.
- We make version 0 software (prototype) to know what we don't know yet.

GitHub repository

ase420/module1/code/2. APIEC/v0-prototype

prototype-v1.py

- Marty already made a prototype.
- Marty does not know software design, so he made code with some smell.

```
names = ["Vera", "Chuck", "Samantha",  
         "Roberto", "Joe", "Dave", "Tina"]  
salaries = [2000, 1800, 1800, 2100,  
            2000, 2200, 2300]  
  
for n in names: print(n)  
for s in salaries: print(s)
```



Vera
Chuck
Samantha
Roberto
Joe
Dave
Tina
2000
1800
1800
2100
2000
2200
2300

- It's good to have a prototype.
- But, something is not right.

- Marty already identified the problem in the code.

Dear Uncle,

this shows a list of your employees. And then it shows a list of all the salaries. Unfortunately I don't know how to connect the names and salaries and I'm late for school.

Your nephew,
Marty

Code smell

- Names and salaries are printed separately.
There is no connection between the employee and their salary!

prototype-v2.py - Fixing the Connection

- We can address Marty's problem using the index.

```
count = len(names)
for c in range(count):
    print(f"{names[c]}, ${salaries[c]}")
```

```
Vera, $2000
Chuck, $1800
Samantha, $1800
Roberto, $2100
Joe, $2000
Dave, $2200
Tina, $2300
```

We Hire Ringo, then What?

- We hire a new employee, Ringo, so we need to **change** our software.
- But we forgot to add Ringo's salary to the salaries list.



Ringo

```
# Added Ringo
names = ["Vera", "Chuck", "Samantha", "Roberto",
         "Joe", "Dave", "Tina", "Ringo"]
# OOPS! Forgot to update salaries
salaries = [2000, 1800, 1800, 2100,
            2000, 2200, 2300]

count = len(names)
for c in range(count):
    print(f"{names[c]}, ${salaries[c]}")
```

| prototype-v3-bug.py - The Error

- And we get an error message.
- Maybe we can easily fix the problem from the error message.

```
Traceback (most recent call last):  
  File "/code/prototype/v3-bug.py", line 6, in <module>  
    print (f" {names [c]}, ${salaries[c]}")  
IndexError: list index out of range
```

Python - The pitfall

- Some static language compilers could find this error when we compile the code.
- Unfortunately, Python is a dynamic language; it does not execute that checking.
- We can find the error only when we run the code.

IMPORTANT

- **Compile-time errors:** The compiler catches these before the code ships to users.
- **Run-time errors:** Users find these while running the code.

It's best to catch errors at compile time, not run time — users don't want to see error messages.

Unlike Java, Python and JavaScript do not require static type checking before execution, so many type-related errors are discovered only at runtime, unless separate static-analysis tools (like `mypy`) are used.

prototype-v3-bugfix.py

We add Ringo's salary to the list and solve this issue.

```
names = ["Vera", "Chuck", "Samantha", "Roberto",  
         "Joe", "Dave", "Tina", "Ringo"]  
salaries = [2000, 1800, 1800, 2100,  
            2000, 2200, 2300, 1900]  
  
count = len(names)  
for c in range(count):  
    print(f"{names[c]}, ${salaries[c]}")
```

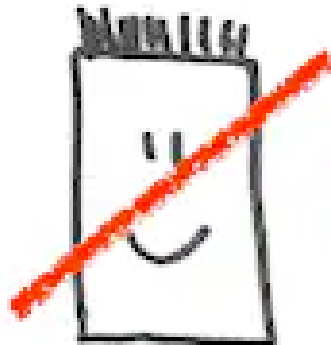
Vera, \$2000
Chuck, \$1800
Samantha, \$1800
Roberto, \$2100
Joe, \$2000
Dave, \$2200
Tina, \$2300
Ringo, \$1900

| The Same Change Problem, Again

- Adding Ringo required us to update two separate lists.
- The next change exposes the same design weakness in a more dangerous way.

Joe Quits, then What?

- Joe quits.
- So, we need to **change** our software again.
- But again, we forgot to delete Joe's salary.



Joe

```
names = ["Vera", "Chuck", "Samantha", "Roberto",  
         "Dave", "Tina", "Ringo"]  
# OOPS! Forgot to delete Joe's salary  
salaries = [2000, 1800, 1800, 2100,  
            2000, 2200, 2300, 1900]  
  
count = len(names)  
for c in range(count):  
    print(f"{names[c]}, ${salaries[c]}")
```

Code Smell and the Rule of Three

- When one change impacts multiple parts of the code, it is a code smell.
- We have now seen the same maintenance problem more than once.
- If it happens a third time, the **Rule of Three** tells us to redesign.

prototype-v4-bug.py - Silent Bug

- When we check the output, we find something wrong.
- Problem: Dave shows Joe's salary (\$2000), Tina shows Dave's salary (\$2200), etc.

- We forgot to delete Joe from the salary list.
- This bug is **harder to find** than before because Python did not print out any error messages.

Vera, \$2000	Vera, \$2000
Chuck, \$1800	Chuck, \$1800
Samantha, \$1800	Samantha, \$1800
Roberto, \$2100	Roberto, \$2100
Joe, \$2000	Dave, \$2000
Dave, \$2200	Tina, \$2200
Tina, \$2300	Ringo, \$2300
Ringo, \$1900	

prototype-v4-bugfix.py

We deleted Joe's salary, and everything worked fine.

```
names = ["Vera", "Chuck", "Samantha", "Roberto",  
         "Dave", "Tina", "Ringo"]  
salaries = [2000, 1800, 1800, 2100,  
            2200, 2300, 1900] # fix
```

Vera, \$2000
Chuck, \$1800
Samantha, \$1800
Roberto, \$2100
Dave, \$2200
Tina, \$2300
Ringo, \$1900

IMPORTANT

Change - the source of complexity

- Even when we finish our implementation, the problem domain keeps changing.
- We should make software in a way that can **meet changes effectively.**

| **Software Design - the tool for changes**

- The current approach with separate lists is error-prone and hard to maintain.
- We need to design software to meet the changes.

More Changes are coming.

- Mr. Star likes the program but wants to add a **job title** to the report.

NAME	SALARY	JOB TITLE
Vera	2000	Manager
Chuck	1800	Attendant
Samantha	1800	Attendant
Roberto	2100	Cook
Dave	2200	Car Repair
Tina	2300	Car Repair
Ringo	1900	Car Repair



```
Vera, $2000, Manager
Chuck, $1800, Attendant
Samantha, $1800, Attendant
Roberto, $2100, Cook
Dave, $2200, Car repair
Tina, $2300, Car repair
Ringo, $1900, Car repair
```